

Zeroual Oussama

Applied Mathematics Student

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Professional Summary

Final-year Applied Mathematics student with experience in mathematical modeling, numerical simulation, data analysis, artificial intelligence, and control systems. Completed internships across industrial research, space technology, water utilities, and banking. Proficient in Python, MATLAB, machine learning, and scientific computing, with an additional academic background in management. Analytical, adaptable, and comfortable working in multidisciplinary teams.

Education

Engineering Degree & Master's in Applied Mathematics

National Higher School of Mathematics, Algiers, Algeria

2022–Present

- Relevant coursework: Numerical Analysis, Differential Equations, Stochastic Processes, Optimization, Control Theory, Fluid Mechanics, Data Analysis, and Computer Vision.

Master's in Management

School of Advanced Commercial Studies, Kolea, Tipaza

2025–Present

- Relevant coursework: Strategic Management, Marketing, and Human Resource Management.

Professional Experience

UAV Control Research Intern

Research Center in Industrial Technologies (CRTI) — Bou Ismail, Tipaza

June 2026–July 2026

- Applied and compared control methods for several types of unmanned aerial vehicles (UAVs).
- Used the PX4 simulation environment to study UAV dynamics and control workflows.

Geospatial AI Intern

Algerian Space Agency (ASAL) — Sidi Abdellah, Algiers

March 2026–April 2026

- Developed a cloud-removal workflow using deep learning to fuse Sentinel-1 and Sentinel-2 satellite imagery.

Control and Simulation Intern

Algiers Water and Sanitation Company (SEAAL) — Kouba, Algiers

December 2025–January 2026

- Gained practical experience with call-center and Hypervision operations for monitoring major water resources.
- Worked with Topkapi and SCADA operational platforms and the ArcGIS geoportal.

Financial Modeling Intern

Local Development Bank (BDL) — Zeralda, Algiers

June 2025–July 2025

- Analyzed financial datasets to prepare management control reports for internal stakeholders.
- Contributed to budget planning and cost-optimization initiatives.

Selected Projects

SAR–Optical Satellite Image Fusion

March 2026–May 2026

Algerian Space Agency (ASAL) Incubator

- Collaborated in a four-person team to remove clouds from optical satellite imagery using paired SAR and optical data from the SEN12MS-CR dataset.
- Documented the methodology, implementation, experimental protocol, and quantitative evaluation of the reconstruction models.

Predictive Analytics and Kaggle Projects

April 2025

- Built statistical and machine learning models to assess sleep quality using physiological data.
- Developed models to analyze fitness data and predict energy expenditure.

Computer Vision and Image Processing

September 2025–June 2026

University Project

- Developed reproducible Python and Jupyter Notebook workflows for image processing and computer vision.

Liquid Flow Modeling in Pipeline Configurations

June 2026

University Project

- Applied conservation laws and the Navier–Stokes equations to fluid flow in different pipeline configurations.
- Analyzed pressure losses and basic stress–strain considerations in engineering flow systems.

Mathematical Modeling of Neuronal Action Potentials

April 2026–May 2026

University Project

- Studied the Hodgkin–Huxley and FitzHugh–Nagumo models, focusing on stability and bifurcation behavior.
- Presented the mathematical formulation, qualitative analysis, and biological interpretation of both models.

Variational and Learning-Based Image Reconstruction

November 2025–December 2025

- Implemented denoising and reconstruction methods using variational regularization, total variation, ISTA, and deep learning in Python/PyTorch.
- Compared classical and learning-based approaches, including BM3D and Swin Transformer models.

Graph Clustering Methods

March 2025

- Delivered an academic presentation on graph clustering techniques and their applications in network analysis.

Certifications & Professional Development

- Introduction to Digital Marketing — HubSpot Academy, 2026
- Supervised Learning with Python — DataCamp, 2026
- Mathurance Hackathon — National Higher School of Mathematics, 2025
- Bank ABC Masterclass — Injaz El Djazair, March 2025

Languages

- Tamazight: Native
- Arabic: Fluent
- English: Upper-intermediate (B2)
- French: Upper-intermediate (B2)

Technical Skills

- **Programming & Scientific Computing:** Python, MATLAB, R, SQL, LaTeX
- **Data Analysis & Machine Learning:** Pandas, NumPy, Matplotlib, Scikit-learn, PyTorch, TensorFlow
- **Computer Vision & Geospatial Tools:** OpenCV, ArcGIS
- **Platforms & Productivity:** Git, GitHub, Linux, VS Code, Power BI, Excel, Canva

Extracurricular Activities

- **Graphic design:** Created visual identities, posters, and layouts for academic projects and events.
- **Video editing:** Produced multimedia content for academic presentations and school events.